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1	2	3	4	TOTAL
3	2	10	10	25

1) $X = \begin{bmatrix}1\\2\\3\end{bmatrix}$ ✗ $\quad y = \begin{bmatrix}3\\5\\8\end{bmatrix}$ ✓ $\quad w = \begin{bmatrix}w_0\\w_1\end{bmatrix}$ ✓

$\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$

a) $(X^T \cdot X)^{-1} \cdot X^T \cdot y = w^*$

$X^{-1} = \frac{1}{Det} \cdot I(X)$

$Det = 0$

$\begin{bmatrix}1\\2\\3\end{bmatrix} \cdot \begin{bmatrix}1&2&3\end{bmatrix} = \begin{bmatrix}1&2&3\\2&4&6\\3&6&9\end{bmatrix}$ ✗

$X \quad X^T \qquad (X \cdot X^T)$

→ Determinant is 0. That's why $X^T X$ is not invertible ✗

So this solution not unique. If det is not zero, $X^T X$ should is invertible and solution is unique.

$y = w_0 + w_1 x^2$

$3 = w_0 + w_1 \cdot 1^2 \Rightarrow 3 = w_0 + w_1$

$5 = w_0 + 4w_1 \Rightarrow 5 = w_0 + 4w_1$

$8 = w_0 + 9w_1 \Rightarrow 8 = w_0 + 9w_1$

$\begin{bmatrix}1&2&3\\2&4&6\\3&6&9\end{bmatrix}^{-1} \cdot \underbrace{\begin{bmatrix}1&2&3\end{bmatrix} \cdot \begin{bmatrix}3\\5\\8\end{bmatrix}}_{37\ (3+24+10)} = w^*$

$\begin{bmatrix}2{,}33\\0{,}66\end{bmatrix}$

$y = \frac{7}{3} + \frac{2}{3}x^2$ ✗

$w_0 = \frac{7}{3} = 2{,}33$

$12 = 4w_0 + 4w_1$

$+ \quad -5 = -w_0 - 4w_1$

$7 = 3w_0$

$3 - \frac{7}{3} = w_1 = \frac{2}{3} = 0{,}66$

$(X^T \cdot X)^{-1} \cdot X^T \cdot y$

2)

	$Y=0$	$Y=1$
X_1	0,4	0,6
X_2	0,4	0,6

(always $X_1 = 0$)

$P(X_1, Y=0) = P(X_1=0 \mid Y=0) = P(Y=0) \cdot P(X_1=0 \mid Y=0)$

$= \frac{2}{5} \cdot 1 = \frac{2}{5} = 0{,}4$ ✓

(always $X_1 = 1$)

$P(X_1, Y=1) = P(X_1=1 \mid Y=1) = P(Y=1) \cdot P(X_1=1 \mid Y=1)$

$= \frac{3}{5} \cdot 1 = \frac{3}{5}\ (0{,}6)$ ✓

$P(X_2, Y=0) = P(X_2=1, Y=0) + P(X_2=0, Y=0)$

$= P(Y=0) \cdot P(X_2=1, Y=0) = \frac{2}{5} \cdot \frac{1}{2} = \frac{1}{5}$

$+ P(Y=0) \cdot P(X_2=0, Y=0) = \frac{2}{5} \cdot \frac{1}{2} = \frac{1}{5}$

$\frac{2}{5}\ (0{,}4)$

$P(X_2, Y=1) = P(X_2=1, Y=1) + P(X_2=0, Y=1)$

$= P(X_2=1, Y=1) \Rightarrow P(Y=1) \cdot P(X_2=1, Y=1) + P(Y=1) \cdot P(X_2=0, Y=1)$

$\rightarrow \left(\frac{3}{5} \cdot \frac{2}{3}\right) + \left(\frac{3}{5} \cdot \frac{1}{3}\right) = \frac{3}{5} = 0{,}6$ ✗

X_1	X_2	Y	
1	1	1	
1	0	1	
1	1	0	x
0	1	1	x
0	1	0	
0	0	1	x
1	0	0	x
0	0	0	

→

1 1 1
1 0 1
0 0 1 x
0 1 1 x
0 1 0
0 0 0
1 0 0 x
1 1 0 x
1 0 0 } duplicate
1 1 0
1 1 1

$X_1 = 0, \quad X_2 = 1$ should be $Y = 0$

$k=1$ meaning when $Y=0$ copy that one time.

$P(Y=0, X_1=0, X_2=1) = P(Y=0, X_1=0) \cdot P(Y=0, X_2=1)$

$= P(Y=0) \cdot P(X_1=0 \mid Y=0) \cdot P(Y=0) \cdot P(X_2=1 \mid Y=0)$

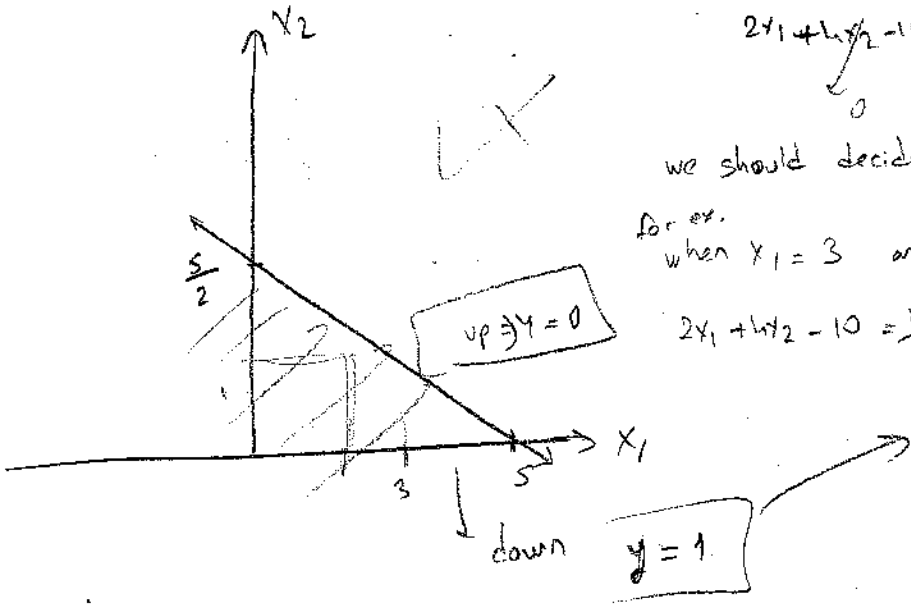
$= \frac{6}{11} \cdot 1 \cdot \frac{6}{11} \cdot \frac{3}{6} = \frac{18}{121}$ ✗

3) $y = \begin{cases} 1, & 2x_1 + 4x_2 - 10 \geq 0 \\ 0, & \text{otw} \end{cases}$

a)

$2x_1 + 4x_2 - 10 = 0$ when $x_1 = 0$: $x_2 = 5/2$

$4x_2 = \frac{10}{4} \frac{5}{2}$ when $x_2 = 0$: $x_1 = 5$



$2x_1 + 4x_2 - 10 = 0$

we should decide where > 0 or < 0

for ex.
when $x_1 = 3$ and $x_2 = 2$

$2x_1 + 4x_2 - 10 \Rightarrow \underbrace{2 \cdot 3}_{6} + \underbrace{2 \cdot 4}_{8} - 10 = 4 > 0$ ✓

b) $P[y \mid x_1 = 2, x_2 = 2] = 1$

$-p \log_2(p) = 1$

$-p(\log_2 \frac{1}{p}) = 0$

$-\frac{1}{2} \cdot \log_2\left(\frac{1}{2}\right) + -\frac{1}{2} \log_2\left(\frac{1}{2}\right)$

-1

$\sigma(1) = \frac{1}{1+e^{-1}} = 0,731$, If $\sigma(z) \geq 0,5$ or $z \geq 0$ this is linear.

So

$\sigma(0) = \frac{1}{1+e^{0}} = 0,5$

4/ a) True. Because when data is excessive and training zero its perfect. (overfitting overload)

b) True

c) False, Because when classification we use another regression

d) True

e) True. Because not lear and no solution